Task 5: Dynamic Quoting Under Inventory Pressure

Approach

The quoting strategy balances spread capture, adverse-selection protection, and inventory risk management. Quoted half-spreads are determined using realized volatility (σ), predicted toxicity (α), inventory (I), and elapsed trading-day fraction (η).

$$Base = (k1 + k2 \cdot \max(0, \alpha - \alpha baseline)) \cdot \sigma$$

$$Skew = Base \cdot (I / ITARGET) \cdot (1 + k4 \cdot \eta^2)$$

An additional end-of-day liquidation term activates during the final 10% of the session to reduce terminal inventory exposure.

$$\delta bid = Base + Skew + Panic$$
$$\delta ask = Base - Skew - Panic$$

All quotes are clipped to the challenge's minimum and maximum spread constraints.

Parameterization Intuition

Volatility determines baseline spread width and compensation for market risk. Toxicity-based widening protects against informed flow by penalizing only unusually toxic client activity relative to each client's historical median toxicity. Inventory skew encourages natural mean reversion by making inventory-reducing trades more attractive.

Inventory effects are normalized using a target inventory level, preventing inventory from overwhelming the quoting logic. End-of-day urgency increases progressively near market close to reduce exposure to terminal inventory penalties.

The baseline spread coefficient was calibrated against the simulator's fill model: $P\_fill = \lambda \exp(-\gamma\delta)$, balancing fill probability and spread capture.

Validation Methodology and Results

The dataset was split chronologically into 60% training, 20% validation, and 20% test periods. Client toxicity baselines were estimated exclusively from training data to avoid look-ahead bias.

Unit tests verified neutrality symmetry, correct inventory-skew direction, toxicity-driven spread widening, and compliance with quoting constraints.

Training: PnL = $318,914.54, Sharpe = 10.91, Max Drawdown = $0.00
Validation: PnL = $96,991.99, Sharpe = 11.09, Max Drawdown = $0.00
Test: PnL = $112,791.54, Sharpe = 11.90, Max Drawdown = $0.00

Performance remains stable across validation and test periods, indicating strong out-of-sample generalization while maintaining effective inventory control and adverse-selection protection.